

Product Requirements Document

ApplyWizz Workday Dry-Run Automation Engine V1

1. Product Name

ApplyWizz Workday Dry-Run Automation Engine

2. One-Line Summary

A controlled automation system that helps ApplyWizz operators apply to Workday jobs using one candidate profile, one resume, one Zoho email account, and a safe human-reviewed dry-run process before final submission.

3. Product Goal

The goal of this product is to automate the most repetitive parts of applying to Workday job links while keeping high-risk answers under human control.

The system should:

- open Workday job links
- create or log in to Workday accounts
- read OTP from Zoho Mail
- upload the candidate's resume
- reach the Workday questionnaire page
- scan questions
- map questions to approved answers
- fill safe answers
- detect conditional questions
- send unknown or risky questions to manual review
- stop before final submission for human approval

The first version should **not** directly auto-submit every application.

The first version should focus on safe automation, visibility, and learning.

4. Background

ApplyWizz helps candidates apply to many jobs through company career portals.

Workday is one of the most common and difficult job application platforms because each company can configure Workday differently.

Problems with Workday:

- different login flows
- account creation required for many companies
- email OTP verification
- resume upload differences
- dynamic questions
- conditional questions
- hidden fields
- dropdowns and radio buttons
- repeated but differently worded questions
- sensitive questions such as visa, sponsorship, disability, gender, salary, relocation, and legal authorization

A simple browser automation script is not enough because Workday forms are dynamic. The system must first understand the questions before filling answers.

5. Core Principle

The product must follow this principle:

Scan first. Understand second. Fill third. Human review if unsure.

The system should never blindly click through the application.

6. Target Users

6.1 Primary User

ApplyWizz internal operator

This user may not be technical. They need a simple dashboard where they can:

- add a candidate
- upload one resume
- add one Zoho email account
- paste Workday job links
- start dry-run automation
- review questions and answers
- approve or reject final submission

6.2 Secondary User

ApplyWizz admin / manager

This user needs visibility into:

- how many jobs were processed
- how many reached questionnaire stage
- how many required manual review
- how many failed
- why they failed
- which Workday companies are working well
- which companies need improvement

6.3 Technical User

Developer / AI coding agent operator

This user needs:

- logs
 - screenshots
 - Playwright traces
 - extracted HTML
 - failed selectors
 - error reports
 - retry information
-

7. V1 Scope

V1 should focus only on **Workday**.

V1 should support:

- one candidate
 - one resume
 - one Zoho email account
 - one or many Workday job links
 - dry-run mode
 - question extraction
 - answer mapping
 - conditional question detection
 - manual review
 - final human approval before submit
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8. Out of Scope for V1

The following should not be built in V1:

- full auto-submit without review
- Greenhouse automation
- Lever automation
- iCIMS automation
- multiple resumes per candidate
- AI resume tailoring
- cover letter generation
- CAPTCHA bypass
- anti-detect browser tools
- proxy rotation
- Browser Use
- Agent Ship
- SkillClaw
- AgentHub
- Alibaba Page Agent
- public candidate dashboard
- payment system
- client-facing reporting

These may be considered later after Workday V1 is stable.

9. Success Definition

V1 is successful if the system can consistently do this:

```
Candidate profile created
↓
One resume uploaded
↓
Workday job link added
↓
Browser opens Workday
↓
Account login or account creation starts
↓
OTP is read from Zoho Mail
↓
Resume is uploaded
↓
Question page is reached
↓
```

```
Questions are extracted correctly
↓
Known questions are mapped to saved answers
↓
Safe answers are filled
↓
Unknown questions go to manual review
↓
System stops before final submission
```

The most important V1 milestone is:

One candidate + one resume + one Workday link + reach the question page + extract questions correctly.

10. Key Metrics

10.1 Automation Metrics

- number of Workday links added
- number of links opened successfully
- number of accounts created successfully
- number of logins completed successfully
- number of OTPs read successfully
- number of resumes uploaded successfully
- number of forms reaching questionnaire page
- number of questions extracted
- number of questions mapped automatically
- number of questions sent to manual review
- number of dry-runs completed
- number of applications approved by human

10.2 Quality Metrics

- question extraction accuracy
- answer mapping accuracy
- conditional question detection rate
- false safe-answer rate
- manual review rate
- failed run rate
- duplicate application detection rate

10.3 Safety Metrics

- number of risky questions blocked
- number of applications stopped before submit

- number of manual overrides
 - number of unknown questions saved to review queue
 - number of final submissions requiring human approval
-

11. User Stories

11.1 Candidate Setup

As an operator, I want to create a candidate profile so that the system knows who is applying.

Required candidate fields:

- full name
 - email
 - phone number
 - location
 - work authorization status
 - target role
 - years of experience
 - LinkedIn URL
 - portfolio URL
 - GitHub URL
 - current company
 - expected salary
 - relocation preference
 - sponsorship requirement
 - availability
-

11.2 Resume Upload

As an operator, I want to upload one resume for the candidate so that the system uses the same resume for all Workday applications.

Rules:

- only one active resume should be used in V1
 - resume should be stored securely
 - resume should be linked to candidate
 - system should use this resume for every job link
 - system should not generate or modify resumes in V1
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11.3 Zoho Email Setup

As an operator, I want to connect a candidate's Zoho email account so that the system can read OTP emails during Workday account creation or login.

Required:

- candidate email
- Zoho OAuth connection or secure credential flow
- OTP email search
- OTP extraction
- OTP logs
- timeout handling

The system should only read OTP-related emails required for automation.

11.4 Job Link Input

As an operator, I want to paste Workday job links so that the system can process them one by one.

Required:

- paste one job link
- paste multiple job links
- validate link format
- detect duplicate links
- store company name if available
- store job title if available
- assign job status

Possible job statuses:

- queued
 - running
 - reached_login
 - otp_required
 - resume_uploaded
 - questionnaire_reached
 - manual_review
 - dry_run_complete
 - approved_for_submit
 - submitted
 - failed
 - duplicate
 - skipped
-

11.5 Workday Browser Automation

As an operator, I want the system to open Workday automatically so that I do not need to manually navigate each job.

The system should use Playwright to:

- open job URL
 - identify Workday page state
 - accept basic cookie banners if required
 - click apply button
 - detect login page
 - detect account creation page
 - detect resume upload page
 - detect questionnaire page
 - capture screenshots at every major step
-

11.6 Workday Account Creation

As an operator, I want the system to create a Workday account when required.

The system should:

- use candidate email
- create password securely
- store account details securely
- detect if account already exists
- handle email verification
- read OTP from Zoho
- continue after verification

The system should never expose passwords in the dashboard.

11.7 Workday Login

As an operator, I want the system to log in to existing Workday accounts.

The system should:

- check if credentials already exist for a Workday tenant
- use stored credentials
- detect incorrect password
- detect OTP requirement
- handle password reset only in future version
- save login success or failure

11.8 Resume Upload

As an operator, I want the system to upload the candidate resume to Workday.

The system should:

- find upload field
 - upload the active resume
 - confirm file appears on page
 - continue to next step
 - save screenshot after upload
 - log upload success or failure
-

11.9 Question Extraction

As an operator, I want the system to scan the Workday question page before filling anything.

The system should:

- capture current page HTML
- compress unnecessary HTML
- send structured HTML to ScrapeGraphAI
- extract questions
- extract field type
- extract available options
- extract required or optional status
- extract visible labels
- extract section name
- extract page step

Question types:

- text
- number
- email
- phone
- date
- dropdown
- radio
- checkbox
- multi-select
- yes/no
- file upload
- address
- legal acknowledgment

11.10 Answer Mapping

As an operator, I want the system to map extracted questions to approved saved answers.

The system should check:

- exact question match
- similar wording match
- known Workday question type
- candidate profile field
- answer bank entry
- risk category

The system can use OpenAI for semantic matching, but only with approved candidate data and saved answer bank.

The system should return:

- question
- suggested answer
- confidence score
- source of answer
- risk level
- whether human review is required

11.11 Safe Answer Filling

As an operator, I want the system to fill only safe answers automatically.

Safe answers may include:

- name
- phone
- email
- address
- LinkedIn
- portfolio
- GitHub
- education
- experience
- employment history
- resume-related fields

Risky answers should require review.

Risky questions include:

- visa sponsorship
- work authorization
- disability
- veteran status
- gender
- race
- ethnicity
- criminal history
- salary expectations
- relocation
- legal certifications
- background check consent

The system should not guess risky answers.

11.12 Conditional Question Detection

As an operator, I want the system to detect new questions that appear after selecting an answer.

Example:

Question: Do you require sponsorship?

Answer: No

New question appears: Are you legally authorized to work in the United States?

The system should:

- fill safe known answer
 - wait for page changes
 - rescan page
 - compare before and after question list
 - extract newly appeared questions
 - map new questions
 - fill if safe
 - send to manual review if unknown
-

11.13 Manual Review

As an operator, I want a manual review screen so that I can approve uncertain questions before submission.

Manual review screen should show:

- candidate name
 - job title
 - company
 - Workday link
 - question
 - field type
 - available options
 - suggested answer
 - confidence score
 - risk level
 - screenshot
 - extracted HTML snippet if needed
 - approve button
 - edit answer button
 - reject button
 - save to answer bank button
-

11.14 Dry-Run Mode

As an operator, I want dry-run mode so that the system can fill the form but stop before final submission.

Dry-run mode should:

- complete all safe steps
- fill known answers
- stop at final review page
- not click final submit
- show a summary to human
- allow approval for final submit later

Dry-run mode is mandatory for V1.

11.15 Final Approval

As an operator, I want to approve the final application before submission.

The system should show:

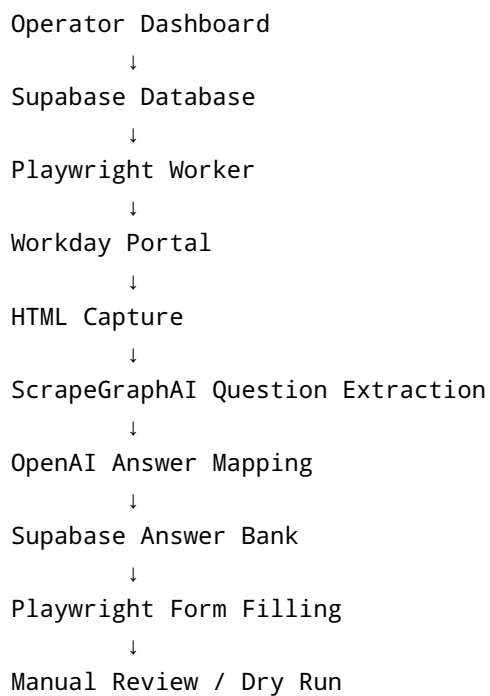
- candidate
- job
- company
- answers filled
- questions skipped

- questions manually reviewed
- warnings
- screenshots
- final submit button

Only after human approval should the system submit the application.

12. System Architecture

V1 architecture:



13. Technology Stack

Frontend

- Next.js
- TypeScript
- Tailwind CSS

Backend

- Next.js API routes
- Node.js worker

- TypeScript

Database

- Supabase PostgreSQL

Storage

- Supabase Storage

Browser Automation

- Playwright

Email OTP

- Zoho Mail API

Question Extraction

- ScrapeGraphAI

Answer Mapping

- OpenAI

Hosting

- Vercel for dashboard
 - Railway, Render, Fly.io, or VPS for worker
-

14. Main Dashboard Pages

14.1 Login Page

Purpose:

Allow ApplyWizz operators to log in securely.

14.2 Dashboard Home

Show:

- total candidates
- total job links
- queued jobs

- running jobs
 - dry-run completed
 - manual review required
 - failed jobs
 - submitted jobs
-

14.3 Candidates Page

Show:

- candidate name
- email
- target role
- resume status
- Zoho status
- total job links
- latest run status

Actions:

- add candidate
 - edit candidate
 - upload resume
 - connect Zoho
 - view job links
-

14.4 Candidate Detail Page

Show:

- profile fields
 - resume
 - Zoho email status
 - answer bank
 - job links
 - application runs
 - manual review items
-

14.5 Job Links Page

Show:

- job URL
- company

- job title
- candidate
- status
- created date
- last run
- error reason

Actions:

- add job link
 - bulk paste links
 - start dry-run
 - retry failed job
 - open Workday link
-

14.6 Runs Page

Show:

- run ID
 - candidate
 - job
 - status
 - started time
 - completed time
 - current step
 - screenshots
 - logs
 - errors
-

14.7 Manual Review Page

Show all questions needing human review.

Actions:

- approve answer
 - edit answer
 - save answer
 - mark unsafe
 - skip job
 - continue run
-

14.8 Answer Bank Page

Show reusable approved answers.

Fields:

- normalized question
 - approved answer
 - category
 - risk level
 - candidate-specific or global
 - last used
 - success count
-

14.9 Settings Page

Show:

- Supabase status
 - Zoho configuration
 - OpenAI key status
 - ScrapeGraphAI key status
 - worker status
 - dry-run mode setting
 - submit approval setting
-

15. Database Requirements

Core tables:

users

Stores internal ApplyWizz operators.

candidates

Stores candidate profile.

candidate_resumes

Stores resume metadata.

zoho_mailboxes

Stores Zoho mailbox connection information.

workday_accounts

Stores Workday tenant account information.

job_links

Stores Workday job links.

application_runs

Stores every automation attempt.

run_steps

Stores each browser automation step.

extracted_questions

Stores scanned questions from Workday.

mapped_answers

Stores suggested and approved answers.

candidate_answers

Stores candidate-specific answer bank.

question_bank

Stores reusable global Workday question patterns.

manual_review_items

Stores questions that need human review.

screenshots

Stores screenshots from each run.

automation_logs

Stores technical logs.

otp_logs

Stores OTP detection and usage logs.

16. Status Flow

Job link status:

```
queued
↓
running
↓
opened
↓
login_required
↓
otp_required
↓
logged_in
↓
resume_uploaded
↓
questionnaire_reached
↓
questions_extracted
↓
answers_mapped
↓
manual_review_required
↓
dry_run_complete
↓
approved_for_submit
↓
submitted
```

Failure statuses:

```
failed_login
failed_otp
failed_resume_upload
failed_question_extraction
failed_answer_mapping
```

```
failed_form_fill
duplicate_application
unsupported_workday_flow
manual_skip
```

17. Risk Rules

The system must classify every question into one of these risk levels:

Low Risk

Can be filled automatically.

Examples:

- name
- email
- phone
- LinkedIn
- GitHub
- portfolio
- address
- education
- work history

Medium Risk

Can be filled only if answer confidence is high and answer exists in approved answer bank.

Examples:

- notice period
- relocation
- expected salary
- preferred location
- availability

High Risk

Must go to manual review.

Examples:

- visa sponsorship
- work authorization

- disability
 - veteran status
 - gender
 - ethnicity
 - criminal history
 - legal certifications
 - background check consent
-

18. Security Requirements

The system must:

- use Supabase RLS where needed
 - store resumes privately
 - store screenshots privately
 - encrypt sensitive credentials
 - never expose service role key to frontend
 - never show raw passwords in UI
 - log access to sensitive records
 - limit OTP email reading to required scope
 - protect candidate personal data
 - avoid storing unnecessary email content
-

19. Logging Requirements

Every automation run should store:

- job link
- candidate
- browser step
- timestamp
- screenshot
- extracted questions
- mapped answers
- confidence score
- errors
- retries
- final status

Logs should be useful for debugging without exposing unnecessary personal data.

20. Error Handling

The system should handle:

- invalid Workday link
- page load timeout
- account already exists
- wrong password
- OTP not received
- OTP expired
- resume upload failed
- question extraction failed
- unknown field type
- unsupported Workday layout
- duplicate application
- network failure
- final submit disabled
- human review required

For each failure, the system should save:

- screenshot
 - error message
 - page URL
 - current step
 - possible reason
 - retry option
-

21. Manual Review Rules

The system should send a question to manual review when:

- confidence score is low
- question is high risk
- answer is missing
- answer options are unclear
- question appears after conditional logic
- multiple possible answers exist
- field type is unsupported
- question wording is new
- OpenAI mapping is uncertain

Manual review decisions should be saved for future reuse only if the operator approves.

22. Dry-Run Summary

At the end of dry-run, show:

- candidate name
- job title
- company
- Workday URL
- account used
- resume uploaded
- total questions found
- total answers filled
- total manual review questions
- high-risk questions
- screenshots
- final page status
- submit readiness score

Submit readiness score:

Ready: all required questions answered, no high-risk unresolved items
Needs Review: some questions require approval
Blocked: missing required answer or unsupported flow
Failed: automation could not complete

23. Version 1 Milestones

Milestone 1: Foundation

Build:

- Next.js app
- Supabase setup
- authentication
- candidate table
- resume upload
- job link table

Success:

Operator can create one candidate and upload one resume.

Milestone 2: Worker Base

Build:

- Playwright worker
- job queue reader
- run status tracking
- screenshots
- logs

Success:

Worker can open a Workday link and save screenshots.

Milestone 3: Login and OTP

Build:

- Workday login/account creation
- Zoho OTP reading
- credential storage

Success:

Worker can log in or create account on at least one Workday tenant.

Milestone 4: Resume Upload

Build:

- resume upload detection
- upload execution
- upload verification

Success:

Worker uploads candidate resume successfully.

Milestone 5: Question Extraction

Build:

- HTML capture
- HTML compression

- ScrapeGraphAI extraction
- extracted question storage

Success:

System reaches questionnaire page and extracts visible questions correctly.

Milestone 6: Answer Mapping

Build:

- candidate answer bank
- global question bank
- OpenAI semantic matching
- confidence score
- risk classification

Success:

System maps common Workday questions to saved answers.

Milestone 7: Form Filling

Build:

- safe field filling
- dropdown handling
- radio handling
- checkbox handling
- text field handling

Success:

System fills low-risk approved answers.

Milestone 8: Conditional Questions

Build:

- after-fill rescan
- before/after question comparison
- new question detection
- conditional question mapping

Success:

System detects and handles newly appearing questions.

Milestone 9: Manual Review

Build:

- review queue
- approve/edit/reject answers
- save answer to bank
- continue run

Success:

Unknown questions can be reviewed by a human.

Milestone 10: Dry-Run Completion

Build:

- stop before submit
- dry-run summary
- readiness score
- final approval screen

Success:

System completes dry-run and waits for human approval.

24. V1 Acceptance Criteria

V1 can be considered complete when:

1. Operator can create a candidate.
2. Operator can upload one resume.
3. Operator can add one Zoho email account.
4. Operator can paste Workday job links.
5. Worker can open Workday link.
6. Worker can log in or create account.
7. Worker can read OTP from Zoho.
8. Worker can upload resume.
9. Worker can reach questionnaire page.

10. Worker can extract visible questions.
 11. System can map known questions to approved answers.
 12. System can identify risky questions.
 13. System can fill safe answers.
 14. System can detect conditional questions.
 15. Unknown questions go to manual review.
 16. System stops before final submit.
 17. Human can approve final submit.
 18. Logs and screenshots are saved for debugging.
-

25. Future Versions

V2

- auto-submit for low-risk applications
- Workday tenant profiles
- better retry logic
- answer learning from manual review
- Playwright trace viewer
- Hermes `/learn` repair system
- advanced duplicate detection

V3

- Greenhouse support
- Lever support
- iCIMS support
- candidate progress reporting
- internal analytics dashboard
- bulk application scheduling

V4

- SkillClaw private engineering knowledge system
 - Agent Ship for multiple internal agents
 - Alibaba Page Agent for dashboard copilot
 - advanced AI operations assistant
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26. Recommended Build Order

The correct first build order is:

1. Dashboard foundation
2. Candidate profile

3. Resume upload
4. Job link input
5. Playwright worker
6. Workday open + screenshots
7. Login/account creation
8. Zoho OTP
9. Resume upload to Workday
10. Reach questionnaire page
11. Extract questions
12. Map answers
13. Fill safe answers
14. Detect conditional questions
15. Manual review
16. Dry-run summary
17. Human approval
18. Controlled submit

27. Final Recommendation

Build the **ApplyWizz Workday Dry-Run Automation Engine V1** first.

Do not build every advanced agent tool immediately.

The first priority is to prove the core Workday flow:

```
One candidate
+
One resume
+
One Zoho email
+
One Workday job link
+
Reach question page
+
Extract questions correctly
```

Once this works reliably, the system can slowly move toward safe filling, conditional question handling, manual review, and later controlled auto-submit.

This approach gives ApplyWizz the safest path to real automation without risking incorrect applications or uncontrolled submissions.